

BUSINESS ANALYST CASE STUDY

Case Study: Hotel Performance Dashboard

1. Business Problem

A hotel has a large amount of booking data, but management does not have a clear view of its **revenue, customer booking patterns, room performance, booking channels, and cancellations**.

Analyzes hotel booking data to understand bookings, revenue, room performance, customer booking patterns, booking channels, cancellations, and monthly trends.

2. Business Objective

The main objectives are:

- To identify the reasons for the decline in sales.
- Analyze total bookings and total revenue.
- Calculate Average Booking Value and Average Stay.
- Monitor cancelled bookings and cancellation rate.
- Compare revenue and bookings across room types.
- Analyze revenue contribution by customer city.
- Compare booking channels using booking volume and revenue.
- Analyze booking status and monthly booking/revenue trends

3. Clarifying Questions

As a Business Analyst, I would ask the following questions:

1. Which room types generate the highest revenue?
2. Which months have the highest and lowest bookings?
3. Which booking channels bring the most customers and revenue?
4. Which cities customers contribute the most bookings and revenue?
5. What is the average length of stay?
6. Which payment methods are most commonly used?
7. What is the average booking value?
8. What specific external or internal factors contributed to the sharp drop in bookings and revenue observed in March?

4. Stakeholders

The important stakeholders are:

- Executive Management (CEO/General Manager): Focused on overall profitability and strategic growth.

- Revenue Management Team: Responsible for pricing strategies and maximizing yield per room.
- Marketing & Sales Team: Needs insights on channel performance and regional demand to optimize campaigns.
- Front Desk & Operations: Deals with day-to-day room allocations, check-ins, and managing no-shows.

5. Data Required

To analyze the problem, I would collect the following data:

- Booking records
- Financial metrics
- Customer Detail Data
- Operational details
- Time series data

The hotel booking dataset may contain:

Booking_ID, Customer_ID, Customer_Name, Booking_Date, Check_In, Check_Out, Room_Type, Room_Number, Adults, Children, Booking_Channel, Payment_Method, Room_Type, Nights, Total_Amount, Booking_Status, and City.

6. Data Analysis

- Overall KPIs: The dashboard reflects 1K total bookings generating ₹14.1M in revenue, with an average booking value of ₹14.11K and an average stay of 4.0 nights.
- Booking Status: A significant portion of bookings are lost. Only 61.7% are successful, while 19.4% are cancelled and 18.9% result in no-shows.
- Revenue by Room Type: Deluxe rooms are the primary revenue driver (₹5.1M), followed by Double (₹4.2M) and Suite (₹3.2M). Single rooms contribute the least (₹1.6M).
- Customer City Performance: Top revenue-generating customer cities are Bengaluru, Mumbai, and Nagpur, though the distribution across the top 10 cities is relatively balanced.
- Channel Performance: Direct digital channels lead the way, with the Website accounting for 34.8% of bookings and Mobile App at 22.6%. Corporate bookings hold the smallest share at 8.7%.
- Monthly Trends: Revenue and bookings peaked in August. However, there was a drastic and anomalous drop in March.

7. Analysis of Possible Reasons

1. High Cancellation Rate:

Based on the patterns observed, the following factors are likely contributing to the high cancellation rate and channel imbalance:

- Overbooking or flexible cancellation policies on the Website and Mobile App channels may make it easy for guests to cancel without penalty.
- Price-sensitive bookings made through Travel Agents and Walk-ins may be more likely to cancel if a cheaper option appears elsewhere.
- Cancellations may occur because of flexible policies, changes in travel plans, payment failures, duplicate bookings, long lead times, price comparisons, or poor communication before arrival. Payment-method patterns can also reflect customer convenience, trust, corporate policies, and the availability of digital payment options.
- Lack of a cancellation deterrent such as a partial deposit or fee could be encouraging casual or speculative bookings.

2. Difference in Room-Type Revenue:

Room types may generate different revenue because of their prices, availability, seasonal demand, upgrades, and customer preferences. A premium room can produce the most revenue because it has a higher rate, even if it has fewer bookings than a lower-priced room.

3. Monthly Changes in Bookings:

Monthly booking changes may be caused by holidays, local events, weather, school breaks, corporate travel, promotions, or changes in room availability and pricing. Website and mobile channels may attract more customers because they offer convenience and direct promotions, while travel agents may contribute bookings where customers need support or package options.

8. Root Cause Analysis

Digging deeper into the top two issues high cancellations and weak revenue conversion on high-volume channels the most probable root causes are:

No cost or commitment at the time of booking: Online channels (Website, Mobile App) likely allow free cancellation, which removes any commitment from the guest and results in speculative bookings that are later cancelled.

Weak follow-up process for pending bookings: Nearly 1 in 5 bookings sits in a pending state, suggesting there is no strong automated reminder or confirmation workflow to convert these into confirmed bookings before they lapse into cancellations.

Inconsistent pricing and discounting across channels: The steep drop in revenue from Website to Corporate suggests channel-specific discounting is not aligned with a common yield management strategy, eroding margins on the same rooms sold through different channels.

And another major reasons of cancellation is uploaded image on mobile app or website is not real images.

Some time price difference on booking by different channel website and walk-in.

9. Recommendations

Based on the analysis, I would recommend:

1. Introduce a partial deposit or stricter cancellation window for Website and Mobile App bookings to reduce speculative cancellations.
2. Set up automated reminders and payment follow-ups for pending bookings within 24-48 hours to reduce the pending-to-cancelled conversion.
3. Standardize pricing and discount rules across all channels so that revenue per booking stays consistent, especially for Travel Agent and Corporate channels.
4. Protect and promote Deluxe and Double room inventory since they are the largest revenue contributors, through better allocation during high-demand periods.
5. We offering them free complimentary services as breakfast, laundry services and pure drinking water.
6. Uploaded real images on mobile app and website this make easy to decide the customer for booking and confirming of booking by calling them or send them message on WhatsApp is very important.
7. Offering more discount to loyal customers and distribute them baches like gold, silver, and platinum give them services according them.
8. Introduce loyalty or repeat-booking incentives on the Website and Mobile App to convert first-time bookers into confirmed, repeat guests.

10. Measuring Business Impact

After implementing the recommendations, the following KPIs can be monitored:

KPI / Metric	Current	Target
Cancellation Rate	19.40%	Below 12%
Average Booking Value	₹14.11K	Increase by 8-10%
Average Stay	4.0 Nights	Improve room utilization
Total Revenue	₹14.1M	Increase by 10-15%
Total Booking	1K	Increase booking volume

11. Final Conclusion

The **Hotel Performance Dashboard** provides a centralized and interactive view of hotel booking and revenue performance. The project transforms raw booking data into meaningful **KPIs, trends, comparisons, and business insights** using data analysis and visualization techniques.

The analysis helps management understand **total bookings, revenue, average booking value, average stay, room-type performance, customer cities, booking channels, and cancellations**. By using interactive filters and detailed analysis, management can identify performance patterns and areas that require further investigation.